

ATTENDANCE MANAGEMENT SYSTEM

Submitted by:

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Technology Used:

Python

Project Type:

Mini Project

1. Introduction

The Attendance Management System is a simple Python-based mini project used to maintain student attendance records. It allows the user to add student names, mark their attendance, and display the attendance report.

2. Objective

The main objectives of this project are:

- To maintain student attendance records.
- To add student names.
- To mark students as Present or Absent.
- To display the attendance report.
- To understand the use of Python programming concepts in a real-world application.

3. Features

The system provides the following features:

- Add student names
- Mark attendance
- Display attendance report
- Simple menu-driven interface
- Easy to use

4. Technologies Used

Programming Language: Python

Tools: VS Code / Python IDLE

Concepts Used:

- Lists
- Dictionaries
- Functions
- Loops
- Conditional statements
- User input

5. System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM

Software Requirements

- Windows/Linux/macOS
- Python 3.x
- VS Code or Python IDLE

6. Algorithm

1. Start the program.
2. Display the main menu.
3. Select the required option.
4. Add student names to the system.
5. Mark attendance for a student.
6. Display the attendance report.
7. Repeat the process until the user selects Exit.
8. Stop the program.

7. Source Code

The source code of the Attendance Management System is developed using Python. The complete Python program will be included in this section.

```
students = []
```

```
def add_student():
```

```
    name = input("Enter student name: ")
```

```
    students.append({"name": name, "attendance": "Absent"})
```

```
    print("Student added successfully!")
```

```
def mark_attendance():
```

```
    name = input("Enter student name: ")
```

```
for student in students:
    if student["name"] == name:
        student["attendance"] = "Present"
        print("Attendance marked successfully!")
        return
```

```
print("Student not found!")
```

```
def display_report():
    print("\n----- Attendance Report -----")
```

```
if len(students) == 0:
```

```
    print("No students added.")
```

```
else:
```

```
    for student in students:
```

```
        print(student["name"], "-", student["attendance"])
```

```
while True:
```

```
    print("\n===== Attendance Management System =====")
```

```
    print("1. Add Student")
```

```
    print("2. Mark Attendance")
```

```
    print("3. Display Report")
```

```
    print("4. Exit")
```

```
choice = input("Enter your choice: ")
```

```
if choice == "1":
```

```
    add_student()

elif choice == "2":
    mark_attendance()

elif choice == "3":
    display_report()

elif choice == "4":
    print("Thank you!")
    break

else:
    print("Invalid choice. Please try again.")
```

8. Output

The output of the Attendance Management System will be shown using screenshots after executing and testing the Python program.

```
===== Attendance Management System =====  
1. Add Student  
2. Mark Attendance  
3. Display Report  
4. Exit  
Enter your choice: 1  
Enter student name: Lavanya  
Student added successfully!
```

```
===== Attendance Management System =====  
1. Add Student  
2. Mark Attendance  
3. Display Report  
4. Exit  
Enter your choice: 3  
  
----- Attendance Report -----  
Lavanya - Present
```

9. Conclusion

The Attendance Management System successfully demonstrates the use of Python programming concepts to maintain student attendance records. It provides a simple and easy way to add students, mark attendance, and display attendance reports.